Page 10-285 Q. 6.59

Suppose $S = \{u_1, u_2\}$ is a basis of V and $T: V \to V$ is defined by $T(u_1) = 3u_1 - 2u_2$ and $T(u_2) = u_1 + 4u_2$. Suppose $S' = \{w_1, w_2\}$ is a basis of V for which $w_1 = u_1 + u_2$ and $w_2 = 2u_1 + 3u_2$.

(a) Find the matrices A and B representing T relative to the bases S and S' respectively.

(b) Find the matrix P such that $B = P^{-1}AP$.

Sol:-

(a) $A = [T]_S$

$T(u_1) = 3u_1 - 2u_2$

$T(u_2) = u_1 + 4u_2$

$\therefore A = [T]_S = \begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix}$

$B = [T]_{S'}$

Now, $T(w_1) = T(u_1 + u_2) = T(u_1) + T(u_2)$

$= 3u_1 - 2u_2 + u_1 + 4u_2$

$= 4u_1 + 2u_2$

$T(w_2) = T(2u_1 + 3u_2)$

$= 2T(u_1) + 3T(u_2)$

$= 2(3u_1 - 2u_2) + 3(u_1 + 4u_2)$

$= 6u_1 - 4u_2 + 3u_1 + 12u_2$

$= 9u_1 + 8u_2$

Let $(a, b) = \alpha w_1 + \beta w_2$

$= \alpha(u_1 + u_2) + \beta(2u_1 + 3u_2)$

$= (\alpha + 2\beta)u_1 + (\alpha + 3\beta)u_2$

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